Swiss bicycle traffic data: a complete guide for BI projects

Switzerland offers some of Europe's richest open bicycle counting datasets, with Zurich, Basel, and St. Gallen providing granular, long-running time series ideal for exploratory analysis. The best data comes from automated counting stations operated by city engineering departments, with Basel's records stretching back to 2000 (Bs) and Zurich's to 2009 (Open Data Switzerland) (Stadt-zuerich) — both downloadable as CSV and updated weekly. Combined with MeteoSwiss hourly weather data, ASTRA geocoded accident records, and federal modal split surveys, these sources support a wide range of analytical angles from weather-cycling correlations to COVID impact assessment. Bern, despite having 15 counting stations, does **not** publish raw counting data as open data. (Stadt Bern) This report catalogs every relevant dataset, its exact download location, and how to use them together.

The three cities with production-ready counting data

The core of any Swiss bicycle BI project will be automated counting station data. Three cities publish rich, machine-readable time series:

Zurich provides the flagship dataset: **15-minute interval** bicycle and pedestrian counts from ~25 cross-sections across the city, collected via induction loops since 2009. (Open Data Switzerland) Each year is a separate CSV file (35–58 MB), with direction-separated counts linked to station IDs. A companion geodataset maps every station with coordinates, correction factors, and active date ranges. The city also publishes an outage log and correction factor PDF to support data quality assessment. (Stadt-zuerich) This dataset is updated weekly and lives at `data.stadt-zuerich.ch/dataset/ted_taz_verkehrszaehlungen_werte_fussgaenger_velo`. (Stadt-zuerich) Key columns include `FK_ZAEHLER` (station ID), timestamp, and directional counts. (Stadt-zuerich +2) An older hourly dataset (2010–2015) exists but has been discontinued. (Stadt-zuerich)

Basel boasts the longest time series — **hourly counts from 2000 to present** (Bs) — and uniquely employs three counting technologies: induction loops, FLIR thermal cameras, and traffic signal (LSA) sensors. All data is downloadable as CSV directly from `data-bs.ch/mobilitaet/`, with per-year files available (e.g., `2020_Velo_Fuss_Count.csv`). (Open Data Switzerland +2) Basel also publishes a curated dataset of **events that influenced bicycle traffic** at counting stations (dataset 100074), which is invaluable for anomaly explanation. (Bs) (Bs) Station locations are in dataset 100038. (Bs) The portal at `data.bs.ch` offers an API returning JSON for the most recent two years of data. (Open Data Switzerland)

St. Gallen publishes hourly counts since 2017 (with an archive back to 2011) via `daten.stadt.sg.ch/explore/dataset/velozahlungen-stadt-stgallen/`. (Sg) A pre-aggregated daily dataset is also available. (Sg) The platform supports CSV/JSON export and has a

REST API. While smaller in scale than Zurich or Basel, St. Gallen adds geographic diversity for cross-city comparison.

City	Portal	Time range	Granularity	Format	Stations
Zurich	data.stadt-zuerich.ch	2009–present	15-minute	CSV	~25
Basel	data.bs.ch	2000–present	Hourly	CSV / API	Multiple
St. Gallen	daten.stadt.sg.ch	2011–present	Hourly + daily	CSV / JSON / API	Several
Bern	—	2015–present	—	PDF only (not open)	15 (internal)

Federal datasets: accidents, modal split, and national routes

ASTRA's geocoded accident database is the definitive source for bicycle safety analysis. Available at `opendata.swiss/de/dataset/strassenverkehrsunfalle-mit-personenschaden`, it contains every police-reported traffic accident in Switzerland since 2011, individually geolocated. Open Data Switzerland Each record includes a boolean field `AccidentInvolvingBicycle` (covering both regular bikes and e-bikes), plus severity category, accident type, road type, and temporal fields (year, month, weekday, hour). stadt-zuerich The Zurich city subset is separately available at `data.stadt-zuerich.ch/dataset/sid_dav_strassenverkehrsunfallorte` in CSV, Parquet, GeoJSON, and GeoPackage formats Stadt-zuerich — notably useful since Parquet loads efficiently in pandas. An interactive map at `map.unfalldaten.ch` provides visual exploration.

The **Mikrozensus Mobilität und Verkehr (MZMV)** is Switzerland's authoritative transport behavior survey, conducted every five years (most recent: 2021, with ~55,000 respondents). It provides the **official modal split figures** including bicycle and e-bike shares, along with trip distances, purposes, vehicle ownership, and attitudes. Summary tables are downloadable from the BFS website; microdata requires an application through FORS. Key 2021 findings: e-bike was the only transport mode with increased usage during the pandemic, and **20% of Swiss households now own an e-bike**. The MZMV's modal split data reveals a striking linguistic divide — cycling share is **8.6%** in German-speaking regions versus just **2.9%** in French-speaking and **2.7%** in Italian-speaking Switzerland. PubMed Central

Additional federal datasets include BFS injured-person statistics by transport mode (1992–present, via STAT-TAB), Open Data Switzerland the Veloland Schweiz national cycling route network Open Data Switzerland (geodata from ASTRA/SchweizMobil), Open Data Switzerland and the BFS Swiss Cities Statistics dataset with commuter modal shares at the municipality level. The

BFU SINUS reports provide annual bicycle/e-bike safety analyses with exportable statistics, though these are analytical products rather than raw data.

Weather, air quality, and COVID data for correlation analysis

MeteoSwiss weather data is the single most impactful complementary dataset. Since May 2025, all MeteoSwiss data is freely available under open government data terms. The SwissMetNet automatic weather stations provide measurements at **10-minute, hourly, daily, monthly, and yearly** granularity (GitHub) (GitHub) from ~260 stations nationwide. Key urban stations align perfectly with bicycle counting cities: Zürich/Fluntern (code: SMA), Basel/Binningen (BAS), and Bern/Zollikofen (BER). Variables include temperature, precipitation, wind speed, relative humidity, sunshine duration, and global radiation. Data is accessible via a STAC API at `data.geo.admin.ch/api/stac/v1/collections/ch.meteoschweiz.ogd-smn` (GitHub) with individual CSV files per station. For a BI project, **hourly weather joined on timestamp with bike counts** is the most straightforward and analytically rewarding correlation.

Air quality data from the NABEL network (16 stations, operated by FOEN and Empa) (Empa) offers half-hourly measurements of NO_2, PM10, PM2.5, O_3, and other pollutants. The data query tool at `bafu.admin.ch/en/data-query-nabel` exports to CSV. (Bundesamt für Umwelt) Urban stations like Zürich-Kaserne and Bern-Bollwerk sit near bicycle counting locations, enabling time-matched analysis. For Zurich specifically, OSTLUFT provides additional coverage with stations like Zürich-Stampfenbachstrasse. (AQICN)

COVID-19 mobility monitoring is available at `opendata.swiss/de/dataset/mobilitaets-monitoring-covid-19` in CSV format, tracking mobility indices by transport mode from March 2020 onward. (Open Data Switzerland) The ETH MOBIS:COVID-19 tracking study adds GPS-based panel data from ~1,600 individuals with a September 2019 baseline, (PubMed Central) showing that cycling mode share surged during lockdowns while public transport collapsed. (PubMed) Official COVID case data (`opendata.swiss/en/dataset/covid-19-schweiz`) enables overlay of pandemic stringency with cycling patterns.

Infrastructure, bike-sharing, and spatial datasets

Zurich publishes a comprehensive **bicycle route network** (Velowegnetz) as GeoJSON/Shapefile including gradient data and one-way restrictions, plus the **Velonetzplanung** dataset tracking implementation progress of the Velostrategie 2030's 100 km priority route network. (Open Data Switzerland) Basel provides everyday cycling routes (dataset 100032), (Bs) touristic routes (100033), one-way streets and danger spots (100036), cycling bans (100035), and bicycle parking with capacity data (100241). (Bs) Zurich's **Zweiradabstellplätze** dataset maps all public bicycle parking locations. (Stadt-zuerich)

For bike-sharing, Zurich's **ZüriVelo/PubliBike station locations** are updated daily in CSV/GeoJSON format at `data.stadt-zuerich.ch/dataset/geo_standorte_zuerivelo`. Open Data Switzerland The SBB-hosted PubliBike Bern availability dataset at `data.sbb.ch/explore/dataset/bike-availability-bern/` recorded station-level availability every 10 minutes. Sbb However, **PubliBike ceased operations in December 2025**, with a successor network launching in spring 2026 EPFL — historical data would need to have been self-collected. The federal **sharedmobility.ch** API (`api.sharedmobility.ch`) aggregates all shared mobility providers GitHub using the GBFS standard GitHub but only provides real-time snapshots with no historical archive.

Population data from BFS STATPOP Open Data Switzerland and swisstopo administrative boundaries (swissBOUNDARIES3D, Medium free since 2021) round out the spatial toolkit, enabling per-capita cycling rate calculations at the municipality level.

Ten analytical angles worth exploring

The combination of long-running counting data, weather records, and accident statistics supports numerous hypotheses a BI project could investigate: Bicycle-data

- **Weather-cycling regression model**: Join MeteoSwiss hourly data with bike counts to quantify temperature sensitivity (literature suggests cycling peaks at ~25°C and declines above), the rain deterrent effect (typically 30–50% drop), and wind impact. This is the most straightforward and rewarding analysis, achieving R^2 values of 0.92–0.94 in comparable studies. MDPI
- **Temporal pattern decomposition**: Identify commute (bimodal: 8 AM + 5 PM peaks) versus recreational (unimodal midday peak) days using clustering or PCA on hourly profiles — following Jake VanderPlas's famous Seattle methodology, directly applicable to Zurich data. Python Data Science Hand...
- **COVID lockdown impact**: Replicate Büchel, Marra & Corman (2022, *Transport Policy*), who used 13 Zurich and 20 Basel counters with Random Forest models. Their key finding — cycling recovered quickly but showed **no lasting increase** without policy measures like pop-up bike lanes — is testable with updated data through 2025. PubMed Central ScienceDirect
- **Seasonal amplitude and e-bike effect**: Quantify the winter-to-summer ratio (typically 3–5x) and test whether this gap narrows in recent years as e-bikes reduce weather deterrence.
- **Bicycle accident hotspot mapping**: Overlay ASTRA's geocoded accident data filtered for `AccidentInvolvingBicycle` with infrastructure data to identify dangerous intersections and road types.

- **Cross-city modal share comparison**: Use MZMV data to compare Zurich (11%), Basel (16%), Bern (15%), Winterthur (15%), and St. Gallen, contextualizing with topography and infrastructure investment levels.
- **Air quality correlation**: Test whether higher cycling volumes at counting stations near NABEL monitors correlate with lower NO_2/PM10, or conversely whether poor air quality deters cyclists.
- **Infrastructure investment and uptake**: Correlate the Velonetzplanung implementation timeline with year-over-year changes at nearby counting stations.
- **Station clustering**: Apply PCA or k-means to station-level hourly profiles to classify stations by usage pattern (commute corridor, recreational route, mixed).
- **Event and anomaly detection**: Use Basel's unique events dataset (100074) to build a framework for detecting and explaining traffic anomalies. bs

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Suggested paper structure for a descriptive BI project

A well-scoped university project should focus on **one city's counting data** (Zurich recommended for richness) with **one or two correlation datasets** (weather is the strongest choice). The following structure balances analytical depth with feasibility:

1. Introduction (1–2 pages): Frame with Swiss cycling policy context — the 2018 constitutional amendment (73.6% approval), Swissinfo Roadmap Velo the 2023 Veloweggesetz, BikerNetzwerk AG and Zurich's Velostrategie 2030. Provelozuerich State 2–3 research questions (e.g., "How do weather conditions affect cycling volumes?" and "Did COVID-19 produce lasting changes in cycling patterns?").

2. Data and methods (2–3 pages): Describe the Zurich 15-minute counting dataset (stations, time range, data quality using the outage log), MeteoSwiss hourly weather data, and optionally ASTRA accident data. Document the Python toolchain: pandas for data wrangling, matplotlib/seaborn for static visualizations, and optionally folium for maps. Explain data cleaning steps (handling outages, applying correction factors, merging on timestamps). Stadt-zuerich

3. Exploratory data analysis (5–7 pages): This is the heart of the project, organized as: (a) data overview and quality assessment, (b) temporal patterns across hours, days, weeks, months, and years using calendar heatmaps and ridgeline plots, (c) weather correlation via scatter plots and regression, (d) spatial comparison across stations using small multiples, and (e) a focused analysis of one special period (COVID lockdown or a major infrastructure change).

4. Key findings and discussion (2–3 pages): Synthesize the 3–5 most interesting findings, compare with Büchel et al. (2022) and Rérat et al. (2022), and discuss implications for Zurich's

cycling targets.

5. Limitations and future work (1 page): Note that induction loops cannot detect carbon-rim wheels (Stadt-zuerich +3) or distinguish regular bikes from e-bikes, that station coverage is sparse relative to the full network, and that correlation does not imply causation for weather-cycling relationships.

The recommended visualization toolkit includes **calendar heatmaps** (daily patterns over months), **ridgeline plots** (hourly distributions across seasons), **correlation matrices** (weather variables vs. counts), **small multiples** (station-by-station comparison), and **folium maps** (station locations colored by volume). The (mobv) GitHub project by Purves, Straumann, and Wood provides open-source code for anomaly-detection heatmaps using Zurich and Basel data that can be directly adapted. (github) (mobv)

Quick-reference dataset table

Dataset	URL	Forma
Zurich bike counts	data.stadt-zuerich.ch/dataset/ted_taz_verkehrszaehlungen_werte_fussgaenger_velo	CSV
Zurich station locations	data.stadt-zuerich.ch/dataset/geo_standorte_der_automatischen_fuss__und_velozaehlungen	GeoJS
Basel bike counts	data.bs.ch/explore/dataset/100013/	CSV /
Basel station locations	data.bs.ch/explore/dataset/100038/	GeoJS
Basel events affecting traffic	data.bs.ch/explore/dataset/100074/	CSV /
St. Gallen bike counts	daten.stadt.sg.ch/explore/dataset/velozahlungen-stadt-stgallen/	CSV / API
ASTRA accidents (national)	opendata.swiss/de/dataset/strassenverkehrsunfalle-mit-personenschaden	Geoda

Dataset	URL	Forma
Zurich accidents	data.stadt-zuerich.ch/dataset/sid_dav_strassenverkehrsunfallorte	CSV / I
MZMV modal split	bfs.admin.ch → Mikrozensus Mobilität und Verkehr	XLS / microc
MeteoSwiss weather	data.geo.admin.ch/api/stac/v1/collections/ch.meteoschweiz.ogd-smn	CSV
NABEL air quality	bafu.admin.ch/en/data-query-nabel	CSV ex
COVID mobility monitoring	opendata.swiss/de/dataset/mobilitaets-monitoring-covid-19	CSV
Zurich bike infrastructure	data.stadt-zuerich.ch/dataset/veloweg	GeoJS SHP
Zurich PubliBike stations	data.stadt-zuerich.ch/dataset/geo_standorte_zuerivelo	GeoJS CSV
Basel bike parking	data.bs.ch/explore/dataset/100241/	CSV / GeoJS
Veloland Schweiz routes	opendata.swiss/en/dataset/langsamverkehr-veloland-schweiz	Geoda WMS

Conclusion

The strongest approach for a Swiss bicycle BI project centers on **Zurich's 15-minute counting data** (the richest and best-documented dataset) merged with **MeteoSwiss hourly weather** from the Zürich/Fluntern station. This pairing alone supports temporal decomposition, weather regression, COVID analysis, and year-over-year trend detection. Adding Basel's data enables cross-city comparison with the advantage of Basel's longer time series (back to 2000) and unique event-correlation dataset. Bs ASTRA accident data adds a safety dimension filterable to bicycle-specific records. Gimi9 The Zurich open data portal also provides starter code templates in Python, starter-code and the `mobv` GitHub project offers directly reusable visualization code for

anomaly detection. github mobv All core datasets are freely available Open Data Switzerland under CC0 or CC-BY licenses, downloadable as CSV, and loadable with a single `pd.read_csv()` call — making the technical barrier to entry minimal and the analytical potential substantial.